

Functional Characterisation of Unmapped Genes in Dairy Slurry Metagenomes Using GO and KEGG Annotation

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Abstract

Dairy slurry tanks serve as significant environmental reservoirs for antimicrobial resistance (AMR), containing diverse resistant bacteria, resistance genes, and residual antibiotics. Although extensive metagenomic sequencing has been conducted on these systems, a substantial proportion of genes remain functionally unannotated, constraining understanding of the metabolic processes that shape these microbial communities. In this study, Gene Ontology (GO) and KEGG pathway annotation were applied to unmapped genes from the Baker et al. (2022) dairy slurry mini-tank metagenomic dataset, analysing four timepoints over seven weeks of controlled storage, to characterise temporal changes in the functional profiles of the slurry microbial community. Annotation rates held consistent and ranged from 18-21% for GO and 33-35% for KEGG, with approximately 60% of genes left unannotated. Analytical methods included hierarchical clustering, which revealed that functional profiles grouped primarily by timepoint rather than by tank identity, suggesting storage duration was the dominant factor in shaping functional community structure. Early samples showed enrichment in growth and biosynthesis-associated pathways, whereas later samples showed elevated regulatory and survival-orientated behaviour, consistent with substrate depletion models. Stochastic processes emerged as a likely driver for late tank divergence. Overall, results offered new insights into the evolving slurry microbiome behaviour.

Introduction

With AMR becoming a global health threat, the importance of studying systems that harbour and spread resistance genes has become increasingly recognised. Among these, dairy slurry has emerged as a critical environmental reservoir for monitoring AMR . Baker et al. (2022) demonstrated this by revealing that these systems serve as both a surveillance and control point for AMR, finding that controlled storage of slurry without fresh inflow significantly reduced resistant populations. Despite the depth of the analysis in this study, a significant proportion of genes remained unmapped, representing an unexplored layer of the slurry microbial community whose metabolic roles and possible contributions to AMR remain unknown. The aim of this project is therefore to apply Gene Ontology (GO) and KEGG pathway annotation to these unmapped genes from the Baker et al. (2022) mini-tank dataset, characterising how the functional profile of the slurry microbial community evolves across storage, and what this reveals about the metabolic processes underpinning microbial adaptation and potential links to the AMR ecology in dairy slurry environments.

Literature review

Antimicrobial Resistance

AMR has emerged as one of the most significant threats to global health, driven in part by the continued use of antibiotics across both clinical and agricultural settings. Livestock productions systems, in particular, have driven this trend due to the extensive use of antibiotics to treat and prevent infectious diseases, thus creating selective pressures that favour resistant bacterial populations (Van Boeckel et al., 2017). Consequently, animal wastes, such as manure and slurry, harbour antimicrobial-resistant bacteria (ARB), resistance genes (ARGs), and residual antibiotics, meaning they act as a critical interface in the wider dissemination of resistance (Lima et al., 2020).

Within agricultural waste systems, slurry storage tanks are critical because they receive continual microbial inputs from livestock waste, while acting as temporary holding environments prior to land application. Therefore these tanks represent key locations where resistant bacteria and resistance genes may persist, decline or be redistributed into the surrounding environments. A study conducted by Baker et al. (2022), supports this, and demonstrated that dairy slurry tanks harboured a stable but diverse resistome shaped by continual waste inflow, microbial turnover and mobile genetic elements. Notably, resistance genes associated with both current and historical farm antibiotic use were detected, together with genes linked to antibiotics not recorded on the farm, therefore suggesting a broader environmental influence on the slurry microbiome. The study concluded that slurry tanks not only proved to be a pivotal junction for monitoring AMR but an opportune threshold for controlling its spread, as storage in the absence of fresh inflow significantly reduced resistant bacterial populations before field application.

Baker et al. (2022) established that AMR within these systems is both persistent and diverse. However this represents just one aspect of the slurry environment. Beyond acting as a reservoir for resistance, slurry storage can be characterised as a dynamic microbial ecosystem where broader functional activities and metabolic processes actively evolve over time.

Dairy Slurry Tanks as Dynamic Microbial Ecosystems

Dairy slurry tanks represent complex, dynamic and metabolically active microbial environments, where diverse microbial communities converge and engage in a broad range of interacting biochemical processes. These systems are frequently inoculated with microorganisms from multiple sources, including the rumen, soil, water, and bedding materials (Habtewold et al., 2018). This inflow of microorganisms and organic matter rapidly depletes available oxygen, maintaining a predominately anaerobic environment (Dalby et al., 2021). Within this environment, several specialised bacteria drive diverse functional processes; ranging from the degradation of organic polymers to produce methane (Habtewold et al., 2018) to the transformation of nitrogen compounds through denitrification (Wang et al., 2022).

Another major driver of the dynamism within these microbial ecosystems is the continuous turnover of microbial populations within the slurry tank, where microbial death is balanced by the fresh input of new

microorganisms from incoming slurry (Baker et al. 2022). Additionally, not all of these fresh microbes entering the tank are able to persist, as rumen-derived species may be unable to establish under the specific temperature and substrate conditions of slurry storage (Habtewold et al., 2018), resulting in a community that is constantly shifting in composition.

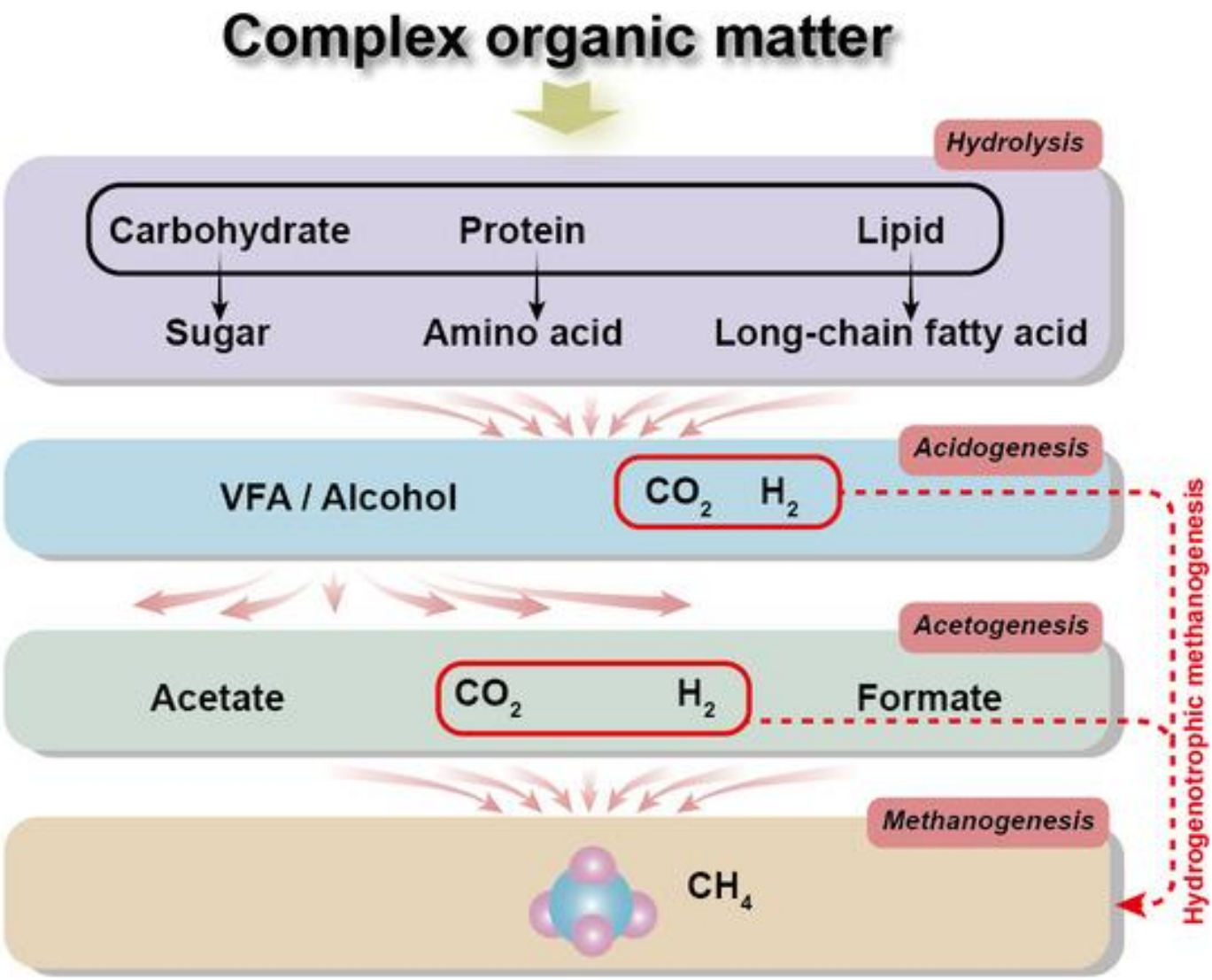
These dynamic microbial communities are sustained by a range of metabolic functions that underpin survival, nutrient cycling and adaptation under anaerobic conditions.

Metabolic Functions in Anaerobic Slurry Environments

The metabolic profile of dairy slurry is strongly shaped by its substantial load of degradable organic matter and the oxygen-deprived conditions that promote anaerobic microbial decomposition during storage (Amon et al., 2006). Within this system, complex carbohydrates, lipids and proteins undergo extracellular hydrolysis, generating simpler soluble compounds including monosaccharides, fatty acids and amino acids that are readily assimilated by microbial populations (Cao et al., 2025). During the early stages of storage, these abundant substrates are likely to select for fast growing heterotrophic taxa, increasing the representation of functions associated with carbohydrate utilisation, amino acid biosynthesis and nucleotide metabolism. This phenomenon has been observed in nutrient rich anaerobic microbial systems, where high substrate availability promotes rapid resource allocation toward biosynthesis and translation (Wu et al., 2017). As storage progresses, the availability of these substrates, including volatile fatty acids, are reduced as a result of continued anaerobic turnover (Patni and Jui, 1985). This depletion likely drives a community transition toward slower-growth states characterised by maintenance metabolism and the increased utilisation of less readily degradable organic matter, consistent with broader models of microbial adaptation under nutrient limitation (Gonzalez & Aranda, 2023).

Ultimately, these upstream hydrolytic and fermentative processes may also facilitate methanogenesis, the final stage of anaerobic digestion as illustrated in Figure1.

Figure 1 – simplified overview of anaerobic microbial degradation observed in anaerobic environments (Cao et al., 2025).



Challenge of Unannotated Genes in Metagenomics

Shotgun metagenomics is a technology that allows researches to comprehensively sample all genes from the organisms present in a given complex sample (Illumina.com, 2013), thus revolutionising microbial ecology by enabling culture independent analysis of entire communities. Despite this, complex metagenomes still see a high rate of sequences that remain unannotated, with estimates suggesting that up to 40-60% of the predicted genes in microbial systems lack functional annotation (Vanni et al., 2022).

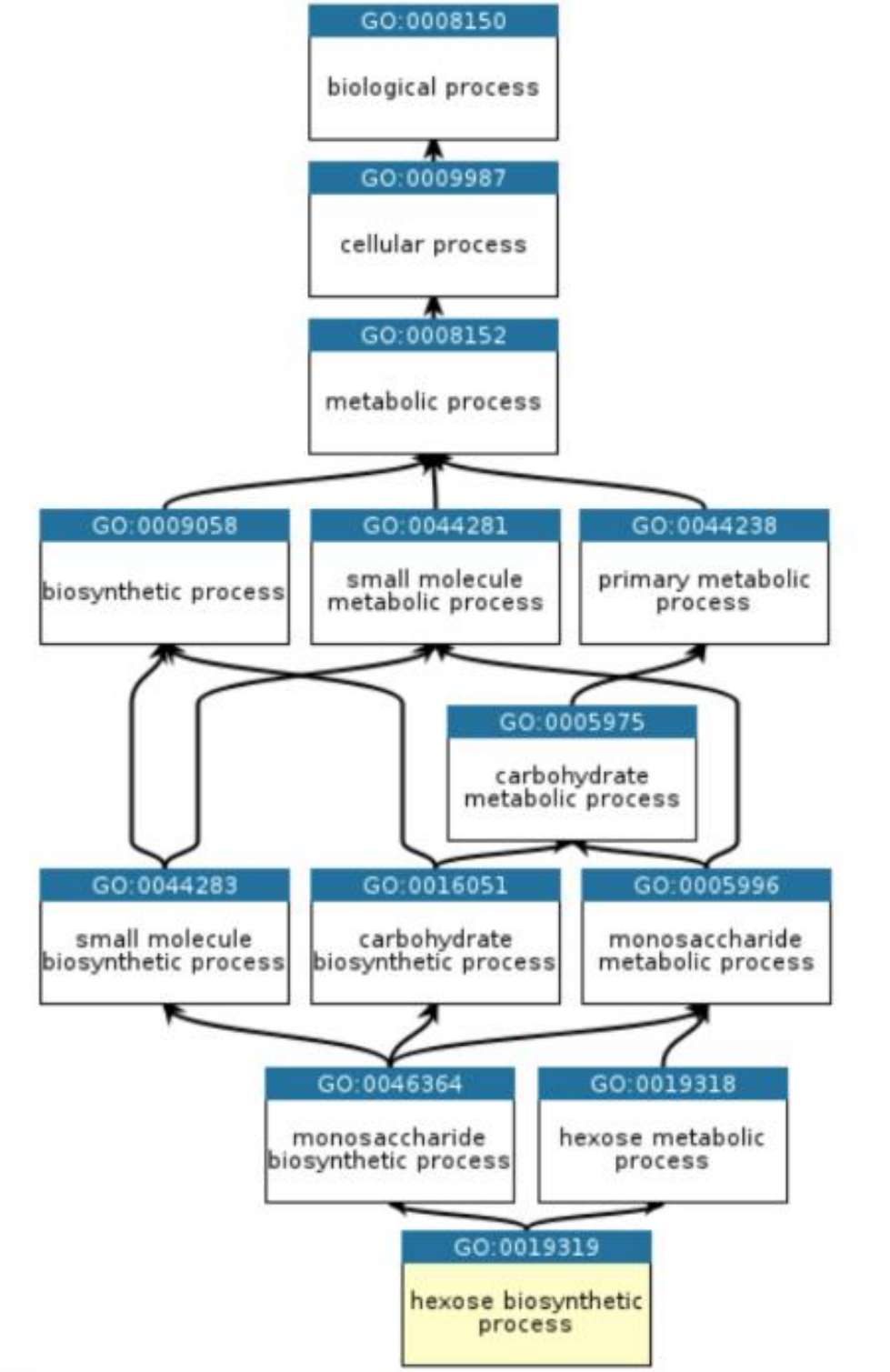
Dairy slurry environments are metabolically diverse and this uncharacterised fraction represents a significant knowledge gap. As unmapped genes could potentially encode novel enzymes responsible for a variety of different processes; for example, those linked to anaerobic metabolism or stress adaptation. This absence therefore prevents researches capturing the full functional landscape of the microbial community, meaning our understanding of the slurry microbiome is only a fragment of its true biological potential.

Gene Ontology as a Framework for Functional Annotation

Despite this limitation, Gene Ontology (GO) remains a powerful tool for inferring gene function in complex metagenomic datasets. GO is a structured framework that describes gene products in terms of molecular function, biological process and cellular component (Thomas, 2016). This framework is particularly useful because instead of just assigning descriptive labels, GO groups genes into hierarchically organised functional categories where a specific child term is positioned under several broader, high-level parent terms (Figure2). This structure is very valuable in metagenomics, because it allows researchers to group diverse gene sequences into high level functional groups, ensuring that large datasets can be interpreted at a broad systems level rather than at the level of individual genes Gene Ontology Resource, n.d.).

GO excels at quantifying how the microbial community's metabolic functional profile changes over time. By comparing the distribution and relative abundance of GO terms across samples (Sutherland et al. 2022), researchers are able to detect functional transitions within the slurry ecosystem. GOs accuracy is reinforced as the hierarchical nature of GO ensures that not just broad trends are identified, but also more specific, lower-abundance terms that signal these environmental changes.

Figure 2 – Details how child term "hexose biosynthetic process" (which represents the more specific biological function) is linked to multiple parent terms category (broader biological categories) (Gene Ontology Resource, n.d.)



KEGG Pathway Analysis

While GO categorises functions, the Kyoto Encyclopedia of Genes and Genomes (KEGG) maps genes to specific metabolic or signalling pathways, situating them within broader biological systems (Metwarebio.com, 2026). Its taxonomy informed pathway reconstruction makes it well suited for analysing complex microbial communities, where diverse organisms may contribute to shared, interconnected metabolic networks (Kanehisa et al., 2023)

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Methodology

Dataset Selection and Metagenomic Sequencing

This study used publicly available metagenomic datasets generated by Baker et al. (2022), who investigated AMR dynamics within dairy slurry storage systems. In the original study twelve mini slurry tanks containing 10L grab samples from the surface of the main slurry tank were created on farm under ambient outdoor conditions. Six treatment conditions were tested in duplicate, with tanks sampled at 0, 2, 4 and 7 weeks after filling.

In this study, only the untreated control mini-tanks were selected to examine how microbial communities functional profiles changed during slurry storage, without influence from antibiotic or metal amendments. Eight samples were analysed in total, representing two biological replicate tanks sampled at four timepoints (T0,T2,T4,T7). These samples were labelled according to tank identity and timepoint (e.g MiniTank1_T0).

Baker et al. (2022) sequenced DNA extracted from mini-tank slurry samples using the Illumina NovaSeq platform, generating paired-end metagenomic reads. Raw reads were then quality filtered and assembled individually, with downstream analyses focused on AMR gene detection and microbial taxonomic classification. Resulting sequence datasets were deposited in the European Nucleotide Archive (ENA) under accession PRJEB38990.

Bioinformatic Processing Pipeline

Figure 3 reveals the analytical workflow used in this study.

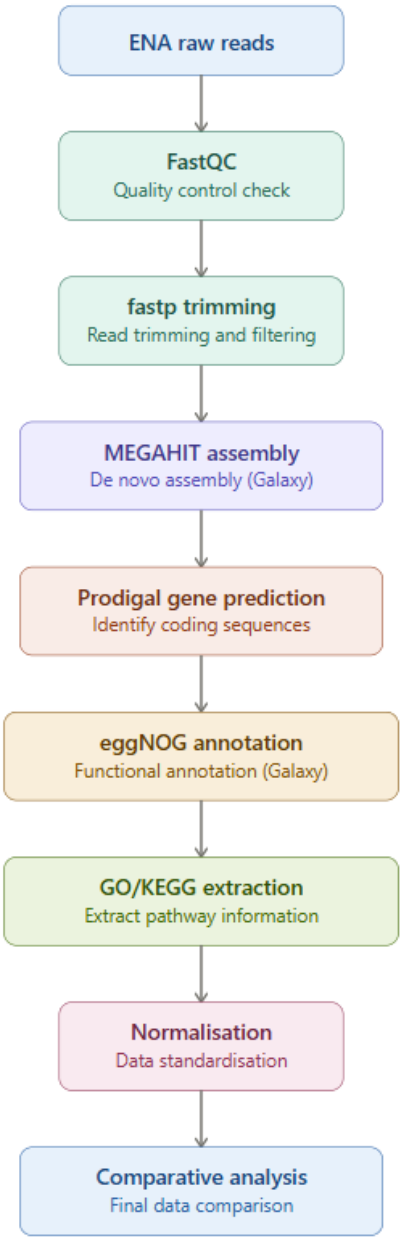


Figure 3 – Overall Project Workflow

Raw paired-end metagenomic reads were downloaded from the ENA and were initially assessed using FastQC v0.11.0 (Andrews, 2010) to evaluate sequence quality, GC content, duplication levels and potential adapter contamination. These reads were subsequently trimmed using fastp v0.23.4 (GitHub, 2022), which removed residual adapters and poor quality reads before downstream assembly.

The resultant high quality reads from each sample were then assembled independently using MEGAHIT v1.2.9 (Usegalaxy.eu, 2026), via the Galaxy Europe platform (due to high computation demands), in meta-sensitive mode (k-mer list: 21,29, 39, 59) to generate contigs.

Prodigal v2.6.3 (Hyatt, 2024) was then used to predict coding regions from assembled contigs, producing translated protein sequences for functional screening.

Protein outputs were annotated using eggNOG-mapper v5.0.2 (Usegalaxy.eu, 2026), also performed via the Galaxy Europe platform, against the eggnog v5 orthology database. The resulting tabular output contained putative gene functions, including Gene Ontology (GO) terms, KEGG Orthology identifiers and KEGG pathway classifications.

Relevant GO and KEGG annotations were then extracted from tabular outputs using Linux command-line tools and converted into per-sample count tables. Where multiple annotations were assigned to a single gene, each annotation was counted independently. Root-level GO terms (GO:0008150, GO:0003674, and GO:0005575) were filtered out prior to downstream comparison.

Total predicted gene numbers varied between samples; therefore annotation counts were normalised to relative abundances (%) to allow direct comparisons across tanks and timepoints. Comparative analyses were subsequently performed to identify dominant functional categories, conserved pathways and temporal shifts during slurry storage

Data visualisation

Final figures were generated in R v4.3.3 using packages including ggplot2, dplyr, tidyr and pheatmap. Visual outputs include bar charts of dominant GO terms and KEGG pathways, temporal line graphs revealing pathway/go term changes across storage stages, and clustered heatmaps.

Clustered heatmaps were constructed using the top 100 most variable GO terms or KEGG pathways across all samples, based on variation in relative abundance values. In both heatmaps, values were row-scaled, meaning colours represented relative abundance within each pathway or GO term across samples. Red indicated relatively higher representation, white intermediate levels and blue relatively lower representation. Hierarchical clustering algorithm was applied to both rows and columns to identify replicate similarity, sample grouping and coordinated temporal changes in predicted functional profiles during slurry storage.

Generative AI tools (ChatGPT v5 and Claude 3.7 Sonnet) were used to assist with selected coding, troubleshooting, and figure-generation tasks.

All KEGG pathways and GO terms were identified directly from their respective online databases: KEGG database (https://www.kegg.jp) and the Gene Ontology Resource (https://www.geneontology.org).

All scripts are available in appendices

Results

Dairy slurry mini-tanks (1&7) were functionally annotated across four timepoints: T0 (week 0), T2 (week 2), T4 (week 4), and T7 (week 7). Open reading frames were predicted from assembled contigs and subsequently annotated using eggNOG-mapper, resulting in GO term and KEGG pathway assignments. Table 1 summarises the total gene counts and annotation rates across all eight samples.

Table 1 – Dataset annotation summary across all mini-tanks and timepoints

Sample	Total Genes	GO Annotated Genes	KEGG Annotated Genes	Unannotated Genes	GO%	KEGG%	Unannotated%
MiniTank1_T0	1,701,866	343,242	591,937	1,003,042	20%	34%	58%
MiniTank7_T0	1,179,839	247,890	413,070	688,489	21%	35%	58%
MiniTank1_T2	1,337,077	253,610	449,705	807,565	18%	33%	60%
MiniTank7_T2	1,285,397	245,849	433,972	773,956	19%	33%	60%
MiniTank1_T4	1,439,565	276,647	486,614	865,604	19%	33%	60%
MiniTank7_T4	1,143,958	221,249	389,448	685,205	19%	34%	59%
MiniTank1_T7	1,508,908	299,202	521,000	893,927	19%	34%	59%
MiniTank7_T7	1,367,131	263,404	463,162	821,419	19%	33%	60%

Table 1 highlights how total gene counts varied considerably across samples, ranging from 1,439,564 (MiniTank7_T4) to 1,701,866 (MiniTank1_T0), thus reflecting the differences in sequencing depth and assembly completeness. Despite this variation, GO annotation rates were highly consistent across all samples ranging from 18-21%. Likewise KEGG pathway annotations rates were similarly stable at 33-35%. Unannotated genes comprised 58-60% of each dataset, a proportion typical of environmental metagenomes, where historically standard orthology-based approaches have yielded low functional coverage in these environments, averaging just 12% for Gene Ontology. Therefore leaving the vast majority of the community uncharacterised as “microbial dark matter” (Maranga et al., 2023).

To account for varying library sizes and total annotation pools, all GO and KEGG counts were normalised to relative abundances (%), calculated as the count of a specific term divided by the total annotation count for that respective database (x100)[1], thus allowing accurate cross-sample comparisons.

Initial Microbial Community State (T0)

(T0), represents the initial functional state of each mini-tank, at establishment, providing a baseline snapshot of the microbial communities and their underlying functional composition at the start of the experiment.

Broad functional overview at baseline

As highlighted in Table 2, both mini-tanks displayed highly similar functional profiles at T0, with the most abundant KEGG pathways and GO terms dominated by high level categories related to central metabolism, biosynthesis and core housekeeping activities, suggesting functional homogeneity between the baseline systems.

Table 2 – Top 5 functional categories in mini-tanks at T0

Rank	KEGG pathway	MiniTank1 (%)	MiniTank7(%)	GO Term	MiniTank1(%)	MiniTank7(%)
1	Metabolic pathways (map01100)	15.82	15.68	Metabolic process (GO:0008152)	1.580	1.559
2	Biosynthesis of secondary metabolites (map01110)	6.59	6.59	Cellular process (GO:0009987)	1.555	1.545
3	Biosynthesis of antibiotics (map01130)	5.19	5.23	Organic substance metabolic process (GO:0071704)	1.507	1.484
4	Microbial metabolism in diverse environments (map01120)	4.41	4.51	Primary metabolic process (GO:0044237)	1.450	1.432
5	Biosynthesis of amino acids (map01230	3.20	3.14	Catalytic activity (GO:0003824)	1.424	1.416

Lower-abundance KEGG Pathway differences

Examination of selected lower-abundance pathways revealed several informative differences between baseline communities (Figure 4). These specific pathways provide greater discriminatory value than broad umbrella categories, as smaller shifts in transport, sensing, and substrate utilisation functions may reflect meaningful variation in microbial community strategy.

Figure 4 – Relative abundance differences in selected KEGG pathways between MiniTank1 and MiniTank7 at T0

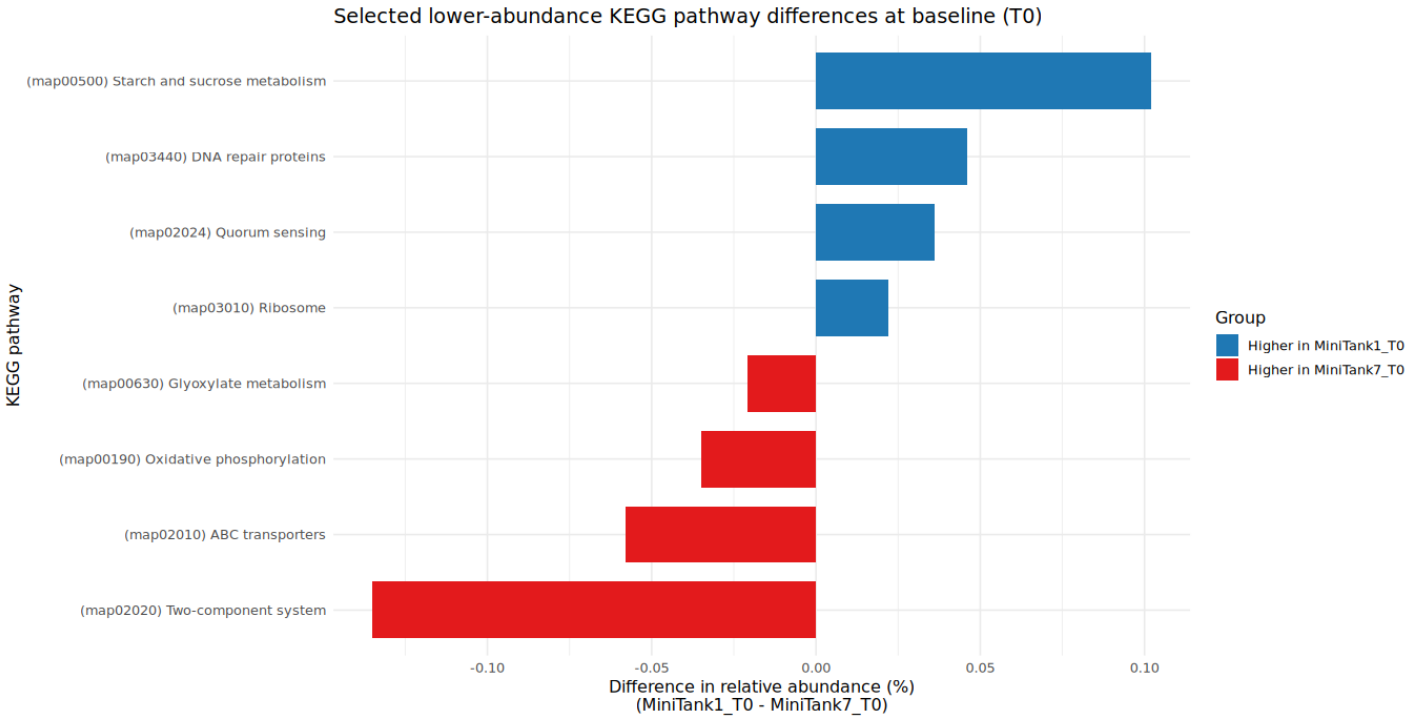


Figure 4 revealed that MiniTank1 showed relatively higher abundance of the following pathways: Starch and sucrose metabolism (map00500), DNA repair proteins (map03440), Quorum sensing (map02024), and Ribosome (map03010). With the most significant difference being in Starch and sucrose metabolism, thus suggesting a stronger representation of carbohydrate degradation and utilisation functions in MiniTank1 at baseline. Additionally, the elevation of ribosome and DNA repair proteins abundance points towards a microbial community with modestly greater translational capacity and stronger genomic maintenance potential.

Conversely, MiniTank7 showed relatively higher abundance of Two-component system (map02020), ABC transporters (map02010), Oxidative phosphorylation (map00190), and Glyoxylate and dicarboxylate metabolism (map00630). With the most pronounced difference overall being the enrichment of the Two-component system. Thus suggesting MiniTank7 having a greater representation of regulatory and environmental sensing pathways. Moreover, increased ABC transporters suggests enhanced nutrient uptake, membrane transport or efflux capacity. While enhanced Oxidative phosphorylation levels indicates slightly greater respiratory energy metabolism.

Overall, these KEGG pathway difference suggest a functional bias in MiniTank1 towards substrate utilisation and growth, whereas MiniTank7 was more heavily invested in transport, environmental sensing and respiratory metabolism.

Lower-abundance GO term differences

A similar pattern was revealed, when comparing selected lower abundance GO terms (Figure5). While broad high level GO terms were nearly identical across both tanks at t0, several lower-level terms unravelled measurable baseline contrasts.

Figure 5 - Relative abundance differences in selected GO terms between MiniTank1 and MiniTank7 at T0

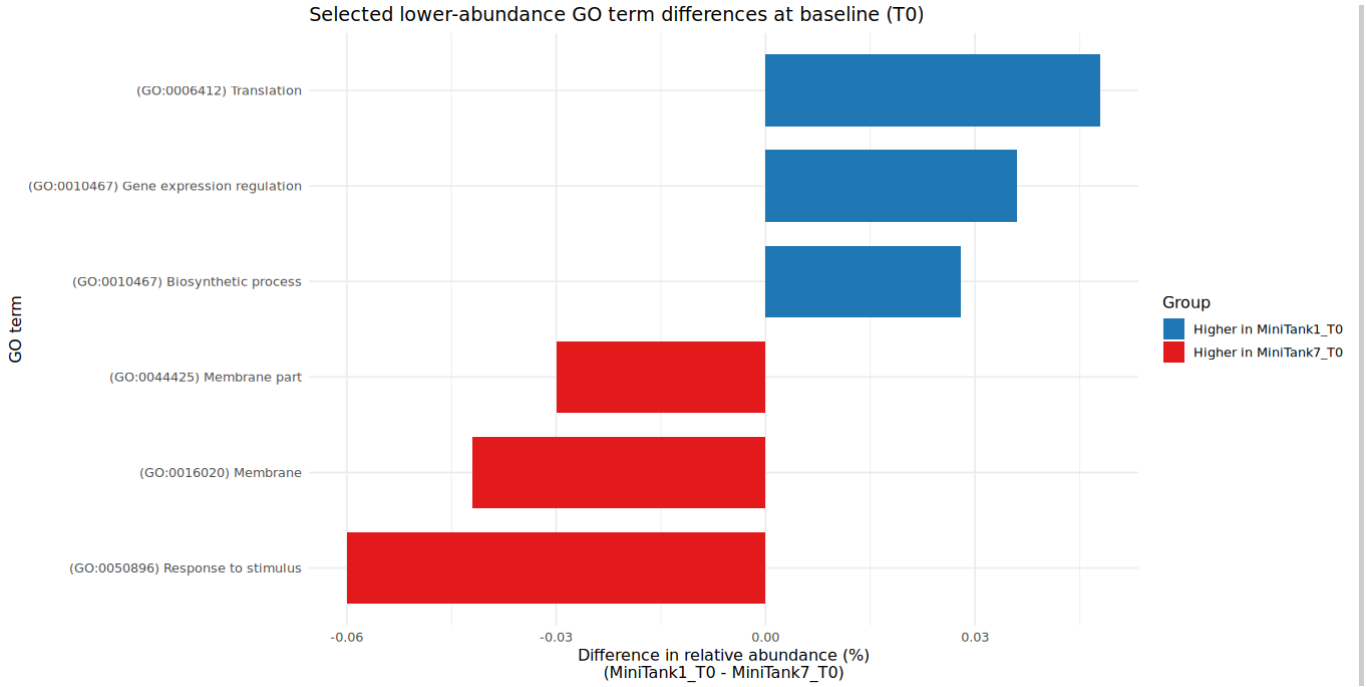


Figure 5 revealed that Translation (GO:0006412) had the largest positive difference between the mini-tanks at (T0), indicating a greater relative level of protein synthesis-related activity in MiniTank1_T0 compared with MiniTank7_T0. Notably, this aligns with the higher relative abundance of Ribosome (map03010) observed in the KEGG comparison (Figure1), providing cross-database validation for a modest increase in translational capacity within MiniTank1 at baseline. Additionally, elevated Gene expression regulation (GO:0010467), and Biosynthetic process (GO:0009058) observed in MiniTank1_T0, further support a microbial community with a functional profile primed for active growth and high biosynthetic turnover.

Additionally, Figure 5 also reveals that Membrane (GO:0016020) and Membrane part (GO:0044425) were elevated in MiniTank7. These categories align with the increased ABC transporters and Two-component systems found in the KEGG analysis, as both functions depend on membrane-bound machinery. Coupled with a significant enrichment in Response to stimulus (GO:0050896), these results reinforce a baseline profile geared toward environmental sensing and transmembrane transport.

Cross-system comparison across all timepoints

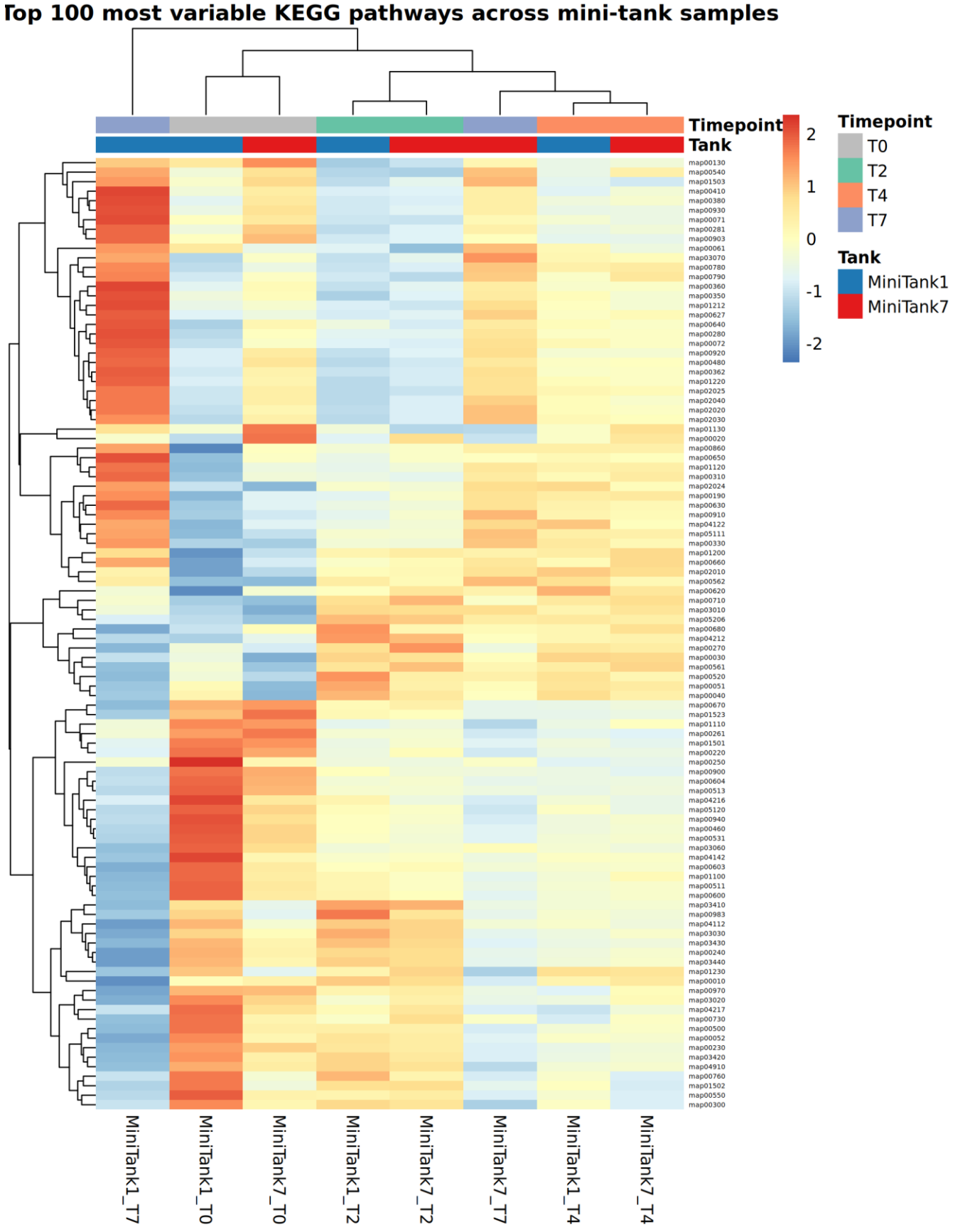
The next stage of analysis examined whether these early contrasts observed persisted through time or whether sampling stage became the stronger driver of community function. To assess this, hierarchical clustering was performed on row-scaled heatmaps using the 100 most variable normalized KEGG pathways and GO terms across all eight samples.

Hierarchical clustering of KEGG pathway profiles

The most significant finding revealed by the KEGG pathway clustering (Figure 6) is that the samples grouped primarily according to timepoint-related functional similarity, rather than by individual tank identity. With the strongest pairings being observed at the intermediate stages; where the mini-tanks clustered closely at T2 and T4, indicating highly similar pathway compositions between systems at weeks 2 and 4.

Furthermore, the earlier observation that the two baseline communities shared a similar initial functional state is reinforced, as they clustered together within the same broader branch. However, clustering at the final timepoint (T7) was weaker. MiniTank7_T7 grouped more closely with the T4 samples, while MiniTank1_T7 formed a distinct branch, indicating greater functional divergence by the end of the experimental period.

Figure 6 - Hierarchical clustering heatmap of the 100 most variable KEGG pathways across all mini-tank samples, showing relative abundance patterns.



Looking past sample clustering, Figure 6 also identifies groups of pathways with shared temporal patterns. For example several pathways, namely: map01100 (metabolic pathways), map00230 (purine metabolism), map00500 (starch and sucrose metabolism), and map01230 (biosynthesis of amino acids), were relatively enriched in early samples (T0-T2). Conversely, later samples (T4-T7), showed a greater relative abundance of map01120 (microbial metabolism in diverse environments), map02020 (two-component system), map02030 (bacterial chemotaxis), map02024 (quorum sensing), and map00190 (oxidative phosphorylation).

Overall, these results reveal that while both mini-tanks followed similar functional trajectories, their pathway compositions were progressively restructured over time.

Hierarchical clustering of GO term profiles

A similar pattern was observed in the GO term heatmap (Figure 7), although clustering was less sharply resolved than for KEGG pathways. Once again tight clustering was observed at T2, indicating strong consistency in ontology level functional composition at this stage.

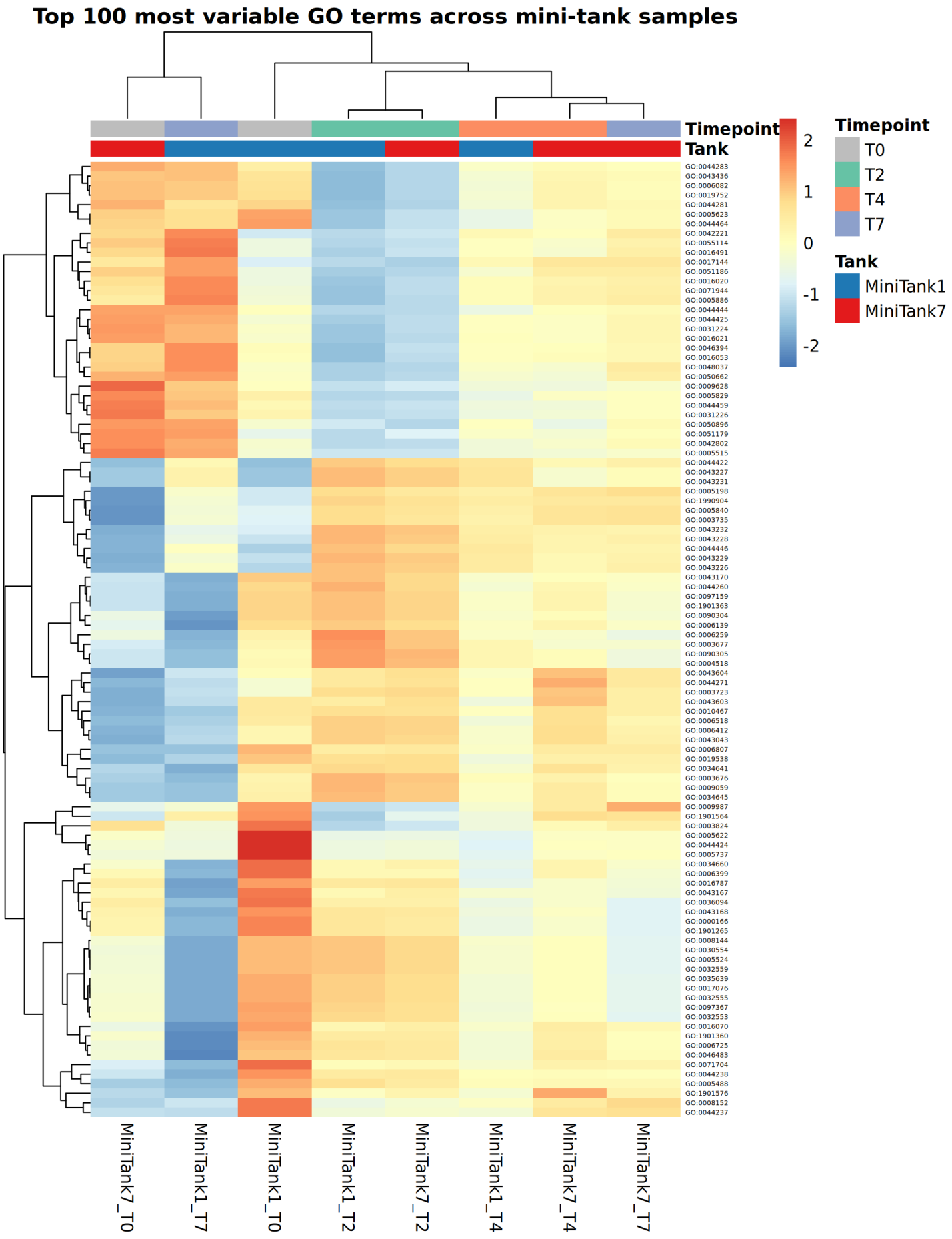
Moreover, Figure 7 showed later-stage samples forming a broader related branch, with MiniTank7_T4 and MiniTank7_T7 clustering closely, while MiniTank1_T4 joined this larger cluster. Interestingly, MiniTank1_T7 shows a more divergent behaviour, remaining distinct from the other late-stage samples, suggesting a shift in microbial community trajectories by the final sampling point.

Individual GO rows also revealed subtle temporal trends across samples. Biosynthesis and small-molecule metabolism related terms, including GO:0043436 (oxoacid metabolic process), and GO:0019752 (carboxylic acid metabolic process), were relatively higher in early samples, T0, but lower in the T2 pair. Potentially revealing that early communities retained a stronger representation of core biosynthetic and substrate-processing functions. Conversely, several broader cellular and regulatory terms, including GO:0055114 (oxidation-reduction process) and GO:0044238 (primary metabolic process), where seen to have a higher expression in later-stage samples, particularly MiniTank7_T4 and MiniTank7_T7.

Notably, MiniTank1_T7 expressed a lower relative abundance across multiple GO rows, including broad categories such as GO:0008152 (metabolic process), contributing to its more distinct clustering position when comparing to other late stage samples.

Looking at both heatmaps together, clustering analysis revealed that samples from equivalent timepoints were generally more similar to one another, compared to samples from different weeks. Concluding that temporal stage exerted a stronger influence on microbial functional structure, than initial differences in baseline communities. The strongest cross system reproducibility was observed at timepoints T2 and T4, where both systems showed near identical functional profiles. However, by T7 the communities began to drift apart, as highlighted by MiniTank1_7, suggesting that later stage community development became less uniform between systems.

Figure 7 - Hierarchical clustering heatmap of the 100 most variable GO terms across all mini-tank samples, showing relative abundance patterns.



Temporal changes within each tank

Specific pathways and GO terms were analysed more closely to examine the temporal shifts within each tank (Figure8/9). This allowed for more specific insight into the **functional profiles of the microbial communities**, which were otherwise masked by dominant high-level terms that remained consistent across all samples.

Figure 8 – Temporal trajectories of selected normalised KEGG pathways across MiniTank1 and MiniTank7 from T0 to T7.

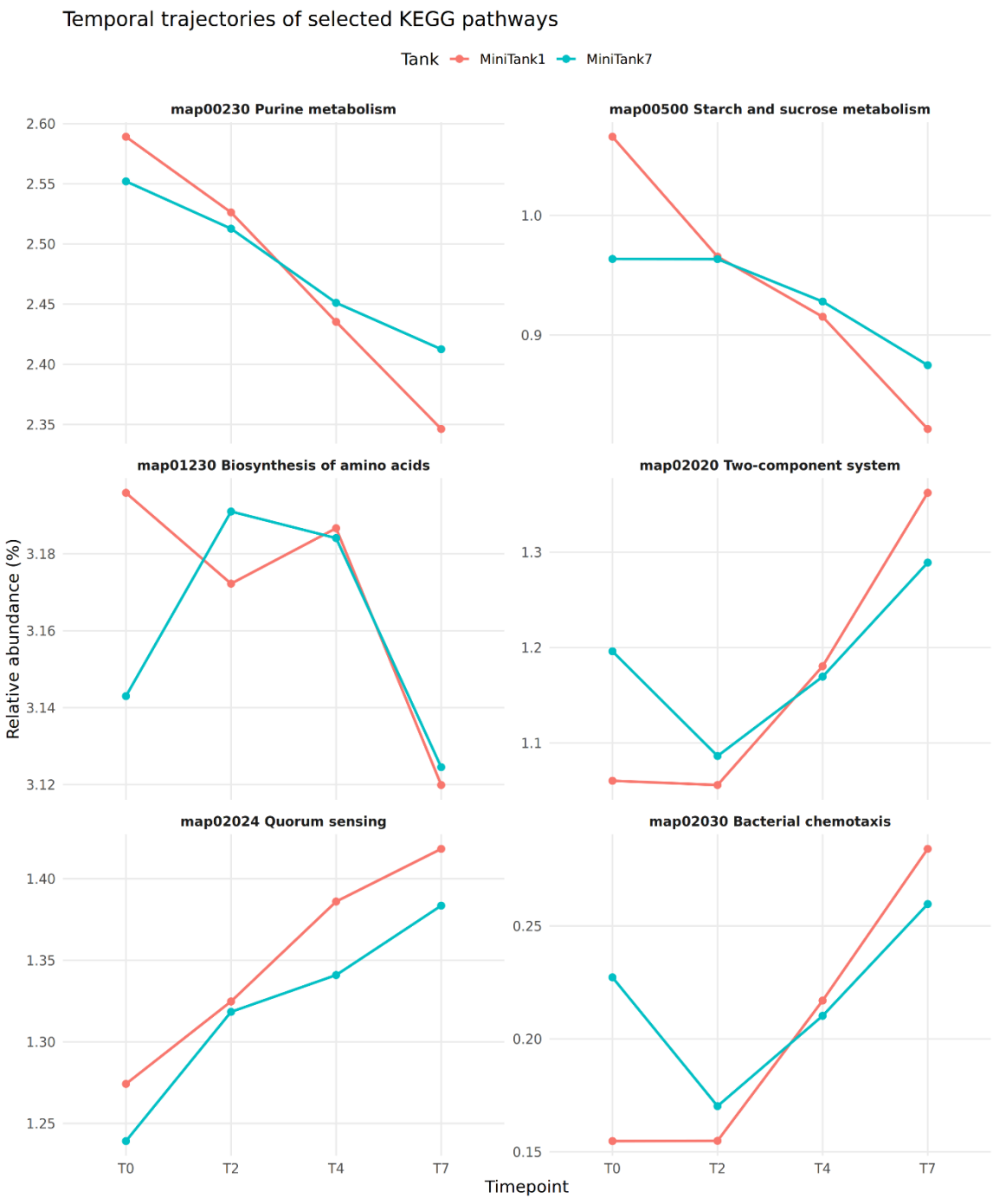
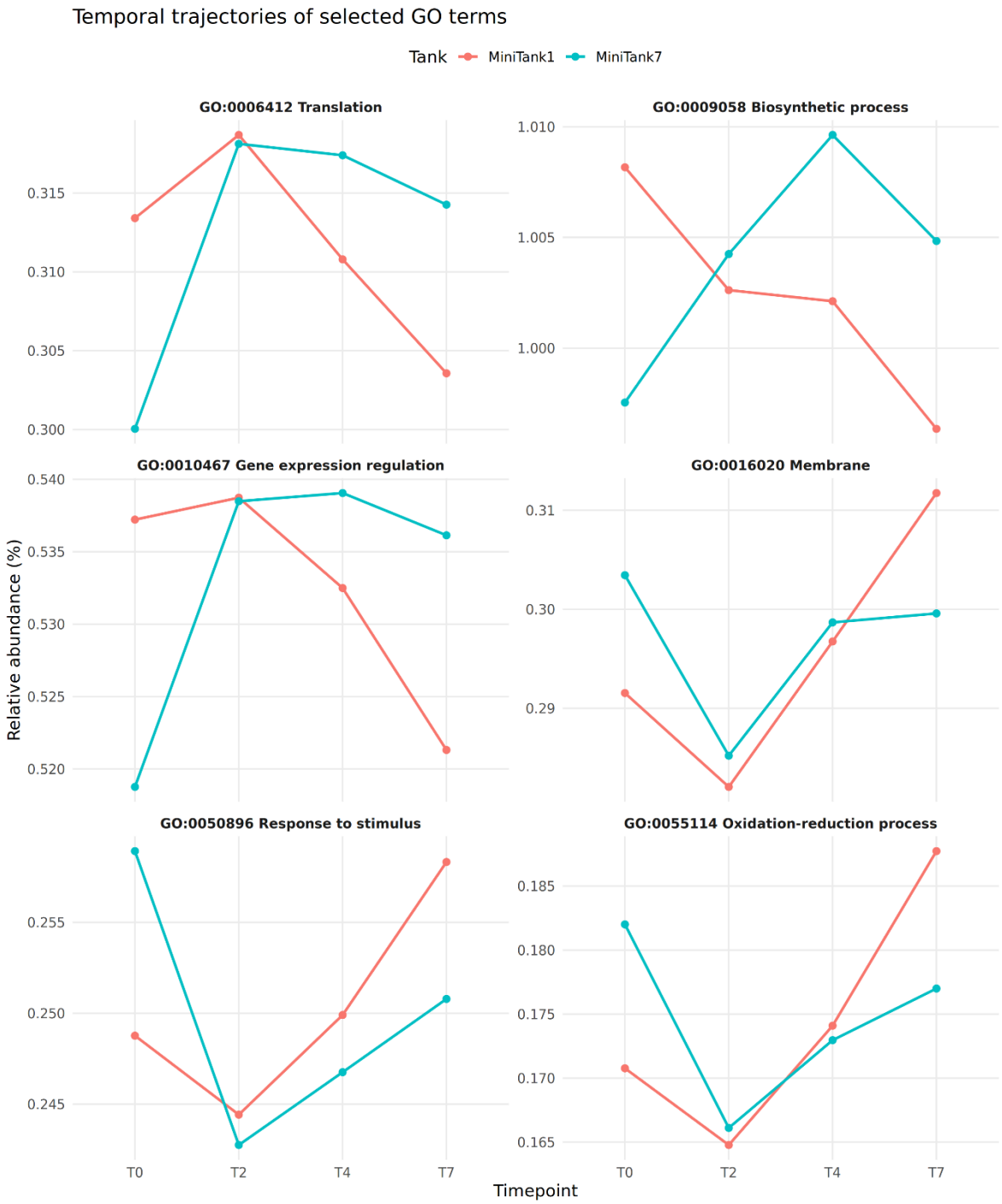


Figure 8 shows consistent temporal trends across both tanks. Growth-associated pathways, including purine metabolism and starch and sucrose metabolism, declined steadily from T0 to T7. Contrarily, adaptive pathways; quorum sensing, two-component system, and bacterial chemotaxis increased, especially at later timepoints. A non-monotonic patter was observed for map01230, peaking mid-experiment before decline at T7.

Figure 9 – Temporal trajectories of selected normalised GO terms across MiniTank1 and MiniTank7 from T0 to T7.

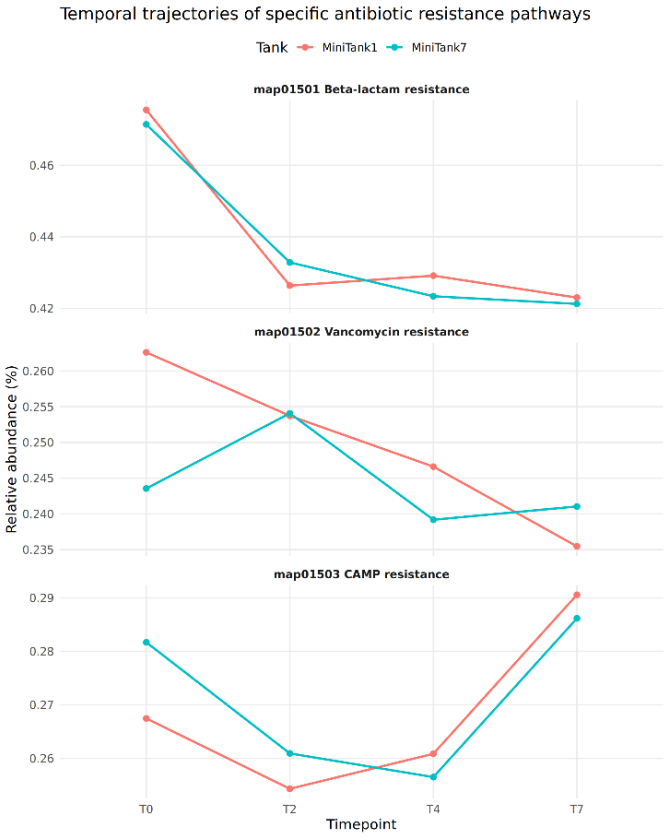


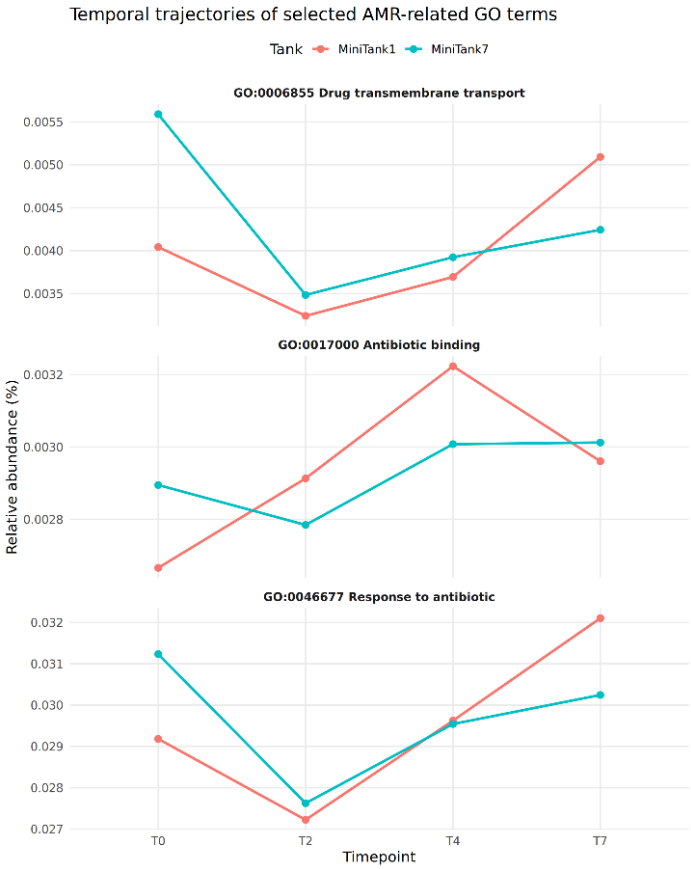
GO term profiles revealed subtle yet consistent temporal restructuring. Growth related terms; Translation and Gene expression regulation were highest at earlier stages and generally decline by T7, with MiniTank1 seeing a steeper decline. Conversely, cellular maintenance and environmental interaction terms, were maintained or increased towards later timepoints.

Both figures indicate a shift from growth associated functions at T0 towards regulatory and adaptive functions by T7.

Resistance-associated functional indicators

The following KEGG pathways and GO terms were used to track AMR potential across all time points.





Discussion

The findings of this study revealed temporally structured functional transitions, subtle baseline differences between tanks and sustained resistance-associated functional capacity during storage

Baseline Communities and Inter-Tank Differences at T0

Both mini-tanks at T0, displayed highly similar high-level functional profiles, with the exact same dominant KEGG pathways and GO terms; reflecting central carbon metabolism, biosynthesis and core housekeeping functions (Table2). This initial functional homogeneity is consistent with the fact that both tanks were filled from the same dairy slurry source, suggesting that they were both inoculated with the same microbial communities.

However, closer examination of lower abundance KEGG pathways and GO terms revealed subtle but informative baseline differences. MiniTank1_T0 showed relatively greater representation of starch and sucrose metabolism (map00500), ribosomal pathways (map03010), and DNA repair functions (map03440), alongside elevated GO terms for translation (GO:0006412) and biosynthetic process (GO:0009058). Altogether, suggesting a microbial community with a functional profile geared towards active growth, high translational activity and carbohydrate substrate utilisation at T0.

On the other hand, MiniTank7_T0 showed enriched levels of the two-component system (map02020), ABC transporters (map02010), oxidative phosphorylation (map00190), and glyoxylate and dicarboxylate metabolism (map00630), alongside elevated GO terms for membrane (GO:0016020) and response to stimulus (GO:0050896). Collectively, these results indicate a microbial community more heavily invested in environmental sensing, membrane-mediated transport, and energetic metabolism at T0. The most notable difference in this sample is the enrichment of two-component signalling systems. These systems represent the cellular signalling circuits which bacteria use to detect and respond to environmental stimuli, including changes in nutrient availability, osmolarity and pH (Mascher, Helmann and Unden, 2006). Therefore, the elevation observed in MiniTank7 at baseline could reflect greater regulatory preparedness within its initial community composition.

Moreover, the alignment between different databases provided important cross-validation, which strengthened the findings. In MiniTank1 elevated ribosomal pathways (KEGG) aligned with increased translation terms (GO). Similarly, in MiniTank7, elevated ABC transporters and two-component systems (KEGG) corresponded with increased membrane and stimulus-response terms (GO). Because both annotation frameworks independently supported the same functional interpretations, confidence in these observations is increased, while ensuring the results were not simply caused by artefacts of either database.

Temporal Functional Restructuring - From Growth to Adaptation

Hierarchical clustering of KEGG and GO profiles across all eight samples revealed a consistent pattern; samples clustered primarily according to timepoint rather than by tank identity. Therefore, indicating that the temporal stage of slurry storage exerted the strongest influence on microbial functional structure. This is one of the key findings of the study, as it revealed how two independent slurry communities can converge, when subjected to the environmental conditions of diminishing substrate availability, accumulating metabolic byproducts, and increasing anaerobic pressure.

The functional structure of early samples (T0-T2) was characterised by greater relative abundance of growth and biosynthesis-associated pathways. Key metabolic signatures included purine metabolism (map00230), starch and sucrose metabolism (map00500), and biosynthesis of amino acids (map01230), as well as GO terms for oxoacid (GO:0043436) and carboxylic acid metabolic processes (GO:0019752). These results are consistent with the high availability of degradable organic substrates expected at T0. Such nutrient rich conditions select for faster growing, more translationally active community members. This aligns with observations in anaerobic systems where increased resource availability drives a community wide increase in ribosomal gene abundance, reflecting a greater allocation of cellular resources toward biosynthesis pathways and translation (Wu et al., 2017). This interpretation is further supported by the taxonomic composition of the source slurry described by Baker et al. (2022), where Firmicutes and Proteobacteria, both phyla associated with fast growing r-selected strategies, were among the dominant groups, consistent with a community primed for rapid growth upon tank establishment.

Conversely, later stage samples (T4-T7) showed a shift toward enrichment of pathways linked with environmental sensing and regulatory adaptation, most notably the two-component system (map02020), quorum sensing (map02024), and bacterial chemotaxis (map02030), alongside increased oxidation-reduction processes (GO:0055114) at the GO level. The enrichment of these pathways towards the end of the storage period reflects microbial strategies witnessed under environmental stress and nutrient limitation.

Quorum sensing is a regulatory system that enables bacteria to monitor cell density and coordinate gene expression through the production of signalling molecules known as autoinducers (AIs). Once these AIs exceed a threshold they bind to cognate receptors and trigger signal transduction cascades. This results in population wide changes in gene expression that control processes such as sporulation, biofilm formation and motility (Ng and Bassler, 2009). These processes collectively represent resource conserving and stress tolerant strategies that confer a survival advantage under nutrient depleted conditions, as observed in late-stage slurry storage. Therefore the increased prevalence of map02024 at T4-T7 suggests a microbial community transitioning towards more collective survival-oriented behaviours.

Similarly, the enrichment of bacterial chemotaxis (map02030) in later samples may reflect increased motility based foraging behaviour as dissolved nutrients become increasingly scarce within the tank. Further supporting the observed shift of the microbial community away from growth-oriented strategies toward resource-seeking and survival orientated behaviour in late stage slurry storage.

Overall, the data revealed a temporal shift from a microbial community focused on growth towards one increasingly represented by regulatory and survival-oriented functions at later time points, driven by the progressive depletion of organic substrates over the storage period, which we know is observed in slurry systems (Patni and Jui, 1985).

Late-Stage Divergence

Despite the close clustering, seen by samples at T2 and T4, by T7 the two tanks began to express differing functional profiles. MiniTank7_7 clustered closely with the T4 samples in the KEGG heatmap, while MiniTank1_T7 formed a more distinct branch. Likewise, in the GO heatmap, MiniTank1_7 expressed lower relative abundance across a variety of broad GO categories, including high level term metabolic process (GO:0008152) which contributed to its distinctly clustered position.

The most plausible driver for this observation is community drift. This is a phenomenon where, after an initial deterministic phase of succession driven by shared environmental filtering, isolated microbial communities begin to diverge progressively because of stochastic processes. Namely; random extinction events, differential colonisation and horizontal gene transfer (Stegen et al., 2012). In relation to this study the subtle tank microenvironmental differences or minor variations in initial inoculum composition, may have become increasing influential, ultimately causing the divergent functional trajectories of the T7 samples.

Resistance Associated Functional Indicators

Relevant KEGG pathways and GO terms were selected to track resistance associated functions across samples. The results revealed that Beta-lactam resistance (map01501) and Vancomycin resistance (map01502) both decreased modestly across the storage duration. While conversely CAMP resistance (map01503) increased at later stages.

The rise of CAMP resistance is particularly interesting, CAMP refers to cationic antimicrobial peptides, which are a group of short positively charged peptides produced by a virtually all organisms as a part of their innate immunity, which function primarily by disrupting bacterial membrane integrity. Bacteria try to resist these CAMPs through multiple mechanisms, including LPS modifications, altering surface charge and efflux pumps (Band and Weiss, 2014). The enrichment of CAMP resistance at later timepoints, therefore suggests that the microbial community has been subject to selective pressures, and as a result began to favour organisms with pre-existing defensive adaptations that allow them to withstand membrane-targeted stressors.

Limitations

Throughout this study, there were several methodological constraints which need to be addressed. Firstly, as Table 1 revealed, 58-60% of all the predicted genes remained unannotated, meaning the functional profile of the microbial community is at best 40% of total gene content. This missing gap represents potentially novel or poor characterised functions, meaning the true functional landscape of the slurry microbiome remains uncharacterised. Moreover, the data only consisted of two replicate tanks this limited statistical power and made it difficult to fully distinguish biological trends from tank-specific variation. Lastly, the results reflect functional potential rather than active gene expression. Therefore, future studies involving metatranscriptomic analysis would be required to confirm whether the enriched pathways were actively transcribed.

Conclusion

Overall, this study revealed that both annotation frameworks offered valuable insights into the changing functions of the dairy slurry microbiome over time. Reflecting a community starting with growth and biosynthesis-related functions in the early stages and shifting to regulatory and survival behaviours later on as substrate availability decreased. Indicators related to resistance also showed changes in the wider functional environment around antimicrobial resistance (AMR). Together, these findings emphasize the importance of functional metagenomics in describing complex environmental microbiomes. They also lay the groundwork for future research that combines metatranscriptomic to better understand the functional aspects of dairy slurry systems.

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Code Appendix

Bioinformatic Pipeline (for simplification code below will be for MiniTank1Timepoint1):

Linux Code to download from ENA:

```
#Mini tank 1 Timepoint 0 (EvalMiniTankPF1)  1

wget -nc ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR810/000/ERR8106110/ERR8106110_1.fastq.gz

wget -nc ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR810/008/ERR8106228/ERR8106228_1.fastq.gz

wget -nc ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR810/000/ERR8106110/ERR8106110_2.fastq.gz

wget -nc ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR810/008/ERR8106228/ERR8106228_2.fastq.gz

wget -nc ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR813/003/ERR8135923/ERR8135923_2.fastq.gz
```

Repeated for all samples

- Mini tank 1 Timepoint 0 (PF1)
    - ERR8106110
    - ERR8106228
- Mini tank 7 Timepoint 0 (PF7)
    - ERR8136035
    - ERR8136039
- Mini tank 1 Timepoint 2 (PF13)
    - ERR8106317
    - ERR8106318
- Mini tank 7 Timepoint 2 (PF19)
    - ERR8106553
    - ERR8106552
- Mini tank 1 Timepoint 4 (PF25)
    - ERR8106787
    - ERR8106786
- Mini tank 7 Timepoint 4 (PF31)
    - ERR8135306
    - ERR8135284
- Mini tank 1 Timepoint 7 (PF37)
    - ERR8135859
    - ERR8135860
- Mini tank 7 Timepoint 7 (PF43)
    - ERR8135923
    - ERR8135926

FastQC Code:

```
fastqc ERR8106110_1.fastq.gz ERR8106110_2.fastq.gz -o fastqc_output
fastqc ERR8106228_1.fastq.gz ERR8106228_2.fastq.gz -o fastqc_output
```

Fastp:

```
#Mini tank 1 Timepoint 0 (EvalMiniTankPF1)  1

fastp \
-i ERR8106228_1.fastq.gz \
-I ERR8106228_2.fastq.gz \
-o trimmed_reads/ERR8106228_R1_trimmed.fastq.gz \
-O trimmed_reads/ERR8106228_R2_trimmed.fastq.gz \
-h trimmed_reads/ERR8106228_fastp.html \
-j trimmed_reads/ERR8106228_fastp.json

fastp \
-w 3 \
-i ERR8106110_1.fastq.gz \
-I ERR8106110_2.fastq.gz \
-o trimmed_reads/ERR8106110_R1_trimmed.fastq.gz \
-O trimmed_reads/ERR8106110_R2_trimmed.fastq.gz \
-h trimmed_reads/ERR8106110_fastp.html \
-j trimmed_reads/ERR8106110_fastp.json
```

MEGAHIT – done via Galaxy Europe (refer to Methodolgy) https://usegalaxy.eu/histories/list

Prodigal

```
MiniTank1_T0

prodigal \

-i MiniTank1_T0_contigs.fasta \

-a MiniTank1_T0_genes.faa \

-d MiniTank1_T0_genes.ffn \

-o MiniTank1_T0_genes.gff \

-p meta
```

eggNOG mapper – also done via Galaxy Europe https://usegalaxy.eu/histories/list

eggNOG code edits after downloaded annotation output

```
1) Code for total genes / % annotated / GO / KEGG

total=$(grep -v "^#" MiniTank1_T0.emapper.tabular | wc -l)

go_count=$(grep -v "^#" MiniTank1_T0.emapper.tabular | awk -F'\t' '$10 != "-" {count++} END {print count+0}')

kegg_path_count=$(grep -v "^#" MiniTank1_T0.emapper.tabular | awk -F'\t' '$13 != "-" {count++} END {print count+0}')

unannotated=$(grep -v "^#" MiniTank1_T0.emapper.tabular | awk -F'\t' '$10 == "-" && $13 == "-" {count++} END {print count+0}')


echo "Total genes: $total"

echo "GO annotated genes: $go_count"

echo "KEGG pathway annotated genes: $kegg_path_count"

echo "Unannotated genes: $unannotated"

echo "Percent GO annotated: $(echo "scale=2; ($go_count/$total)*100" | bc)%"

echo "Percent KEGG pathway annotated: $(echo "scale=2; ($kegg_path_count/$total)*100" | bc)%"

echo "Percent unannotated: $(echo "scale=2; ($unannotated/$total)*100" | bc)%"
```

2) Code to create the clean 5-column annotation file

```
grep -v "^#" MiniTank1_T0.emapper.tabular | cut -f1,8,10,12,13 > MiniTank1_T0_clean.tsv
```

3) Check the clean file has exactly 5 columns

```
awk -F'\t' '{print NF}' MiniTank1_T0_clean.tsv | sort | uniq
```

4) GO term counting code

```
cut -f3 MiniTank1_T0_clean.tsv | grep -v "^-$" | tr ',' '\n' | sort | uniq -c | sort -nr > GO_counts.txt
```

5) KEGG pathway counting code

```
cut -f5 MiniTank1_T0_clean.tsv | grep -v "^-$" | tr ',' '\n' | sort | uniq -c | sort -nr > KEGG_pathway_counts.txt
```

6) Merge KO and Map Ids

```
sed 's/^ *\([0-9]*\) ko/\1 map/' KEGG_pathway_counts.txt > KEGG_fixed.txt
awk '{counts[$2]+=$1} END {for (k in counts) print counts[k], k}' KEGG_fixed.txt | sort -nr > KEGG_merged.txt
```

7) Merge vanity check

```
head KEGG_merged.txt
grep "ko" KEGG_merged.txt
wc -l KEGG_merged.txt
```

8) Go Term filtering

```
grep -v -E "GO:0008150|GO:0003674|GO:0005575" GO_counts.txt > GO_counts_filtered.txt
head -10 GO_counts_filtered.txt > GO_top10.txt
```

9) Normalisation to enable comparison

```
#GO
total=$(awk '{sum += $1} END {print sum}' GO_counts_filtered.txt)
awk -v total=$total '{print ($1/total)*100, $2}' GO_counts_filtered.txt > GO_norm.txt
head -10 GO_norm.txt > GO_top10_norm.txt
#KEGG
total=$(awk '{sum += $1} END {print sum}' KEGG_merged.txt)
awk -v total=$total '{print ($1/total)*100, $2}' KEGG_merged.txt > KEGG_norm.txt
head -10 KEGG_norm.txt > KEGG_top10_norm.txt
```

```
#Kegg Initial Differences R Code
library(ggplot2)
df <- data.frame(
Pathway = c(
"(map02020) Two-component system",
"(map02010) ABC transporters",
"(map00190) Oxidative phosphorylation",
"(map00630) Glyoxylate metabolism",
"(map03010) Ribosome",
"(map02024) Quorum sensing",
"(map03440) DNA repair proteins",
"(map00500) Starch and sucrose metabolism"
),
Diff = c(
-0.135,
-0.058,
-0.035,
-0.021,
0.022,
0.036,
0.046,
0.102
)
)
df$Group <- ifelse(df$Diff > 0,
                   "Higher in MiniTank1_T0",
                   "Higher in MiniTank7_T0")
ggplot(df, aes(x=reorder(Pathway, Diff), y=Diff, fill=Group)) +
geom_col(width=0.75) +
coord_flip() +
scale_fill_manual(values=c(
"Higher in MiniTank1_T0"="#1f78b4",
"Higher in MiniTank7_T0"="#e31a1c"
)) +
labs(
title="Selected lower-abundance KEGG pathway differences at baseline (T0)",
x="KEGG pathway",
y="Difference in relative abundance (%)\n(MiniTank1_T0 - MiniTank7_T0)"
) +
theme_minimal(base_size=12)
```

```
#GO Initial Differences R code

library(ggplot2)


df <- data.frame(

GO = c(

"(GO:0050896) Response to stimulus",

"(GO:0016020) Membrane",

"(GO:0044425) Membrane part",

"(GO:0006412) Translation",

"(GO:0010467) Gene expression regulation",

"(GO:0010467) Biosynthetic process"

),


Diff = c(

-0.060,

-0.042,

-0.030,

0.048,

0.036,

0.028

)

)


df$Group <- ifelse(df$Diff > 0,

                   "Higher in MiniTank1_T0",

                   "Higher in MiniTank7_T0")


ggplot(df, aes(x=reorder(GO, Diff), y=Diff, fill=Group)) +

geom_col(width=0.75) +

coord_flip() +

scale_fill_manual(values=c(

"Higher in MiniTank1_T0"="#1f78b4",

"Higher in MiniTank7_T0"="#e31a1c"

)) +

labs(

title="Selected lower-abundance GO term differences at baseline (T0)",

x="GO term",

y="Difference in relative abundance (%)\n(MiniTank1_T0 - MiniTank7_T0)"

) +

theme_minimal(base_size=12)
```

```
#KEGG Heatmap
ibrary(pheatmap)
files <- c(
  "minitank1/KEGG_norm.txt",
  "minitank2/KEGG_norm.txt",
  "minitank3/KEGG_norm.txt",
  "minitank4/KEGG_norm.txt",
  "minitank5/KEGG_norm.txt",
  "minitank6/KEGG_norm.txt",
  "minitank7/KEGG_norm.txt",
  "minitank8/KEGG_norm.txt"
)
sample_names <- c(
  "MiniTank1_T0",
  "MiniTank7_T0",
  "MiniTank1_T2",
  "MiniTank7_T2",
  "MiniTank1_T4",
  "MiniTank7_T4",
  "MiniTank1_T7",
  "MiniTank7_T7"
)
mat_list <- list()

for (i in seq_along(files)) {
  df <- read.table(files[i], header = FALSE, stringsAsFactors = FALSE)
  colnames(df) <- c("Value", "Pathway")
  df$Value <- as.numeric(df$Value)
  df <- df[!is.na(df$Value), ]
  df <- aggregate(Value ~ Pathway, df, sum)
  mat_list[[i]] <- df
}

all_pathways <- Reduce(union, lapply(mat_list, function(x) x$Pathway))

mat <- matrix(0, nrow = length(all_pathways), ncol = length(mat_list))
rownames(mat) <- all_pathways
colnames(mat) <- sample_names

for (i in seq_along(mat_list)) {
  df <- mat_list[[i]]
  mat[df$Pathway, i] <- df$Value
}
```

```
vars <- apply(mat, 1, var)
vars <- vars[vars > 0]
mat <- mat[names(vars), ]

top100_names <- names(sort(vars, decreasing = TRUE))[1:100]
top100 <- mat[top100_names, ]

annotation_col <- data.frame(
  Tank = c("MiniTank1", "MiniTank7", "MiniTank1", "MiniTank7",
           "MiniTank1", "MiniTank7", "MiniTank1", "MiniTank7"),
  Timepoint = c("T0", "T0", "T2", "T2", "T4", "T4", "T7", "T7")
)

rownames(annotation_col) <- colnames(top100)

annotation_colors <- list(
  Tank = c(MiniTank1 = "#1f78b4", MiniTank7 = "#e31a1c"),
  Timepoint = c(T0 = "#bdbdbd", T2 = "#66c2a5", T4 = "#fc8d62", T7 = "#8da0cb")
)

png("KEGG_top100_heatmap_final.png", width = 2000, height = 2600, res = 250)

pheatmap(
  top100,
  scale = "row",
  fontsize_row = 4,
  fontsize_col = 10,
  border_color = NA,
  annotation_col = annotation_col,
  annotation_colors = annotation_colors,
  main = "Top 100 most variable KEGG pathways across mini-tank samples"
)

dev.off()
```

```
#GO Heatmap
library(pheatmap)
files <- c(
  "minitank1/GO_norm.txt",
  "minitank2/GO_norm.txt",
  "minitank3/GO_norm.txt",
  "minitank4/GO_norm.txt",
  "minitank5/GO_norm.txt",
  "minitank6/GO_norm.txt",
  "minitank7/GO_norm.txt",
  "minitank8/GO_norm.txt"
)
sample_names <- c(
  "MiniTank1_T0",
  "MiniTank7_T0",
  "MiniTank1_T2",
  "MiniTank7_T2",
  "MiniTank1_T4",
  "MiniTank7_T4",
  "MiniTank1_T7",
  "MiniTank7_T7"
)

mat_list <- list()

for (i in seq_along(files)) {
  df <- read.table(files[i], header = FALSE, stringsAsFactors = FALSE)
  colnames(df) <- c("Value", "GO")
  df$Value <- as.numeric(df$Value)
  df <- df[!is.na(df$Value), ]
  df <- aggregate(Value ~ GO, df, sum)
  mat_list[[i]] <- df
}

all_terms <- Reduce(union, lapply(mat_list, function(x) x$GO))

mat <- matrix(0, nrow = length(all_terms), ncol = length(mat_list))
rownames(mat) <- all_terms
colnames(mat) <- sample_names

for (i in seq_along(mat_list)) {
  df <- mat_list[[i]]
  mat[df$GO, i] <- df$Value
}
```

```
vars <- apply(mat, 1, var)
vars <- vars[vars > 0]
mat <- mat[names(vars), ]

top100_names <- names(sort(vars, decreasing = TRUE))[1:100]
top100 <- mat[top100_names, ]

annotation_col <- data.frame(
  Tank = c("MiniTank1", "MiniTank7", "MiniTank1", "MiniTank7",
           "MiniTank1", "MiniTank7", "MiniTank1", "MiniTank7"),
  Timepoint = c("T0", "T0", "T2", "T2", "T4", "T4", "T7", "T7")
)

rownames(annotation_col) <- colnames(top100)

annotation_colors <- list(
  Tank = c(MiniTank1 = "#1f78b4", MiniTank7 = "#e31a1c"),
  Timepoint = c(T0 = "#bdbdbd", T2 = "#66c2a5", T4 = "#fc8d62", T7 = "#8da0cb")
)

png("GO_top100_heatmap_final.png", width = 2000, height = 2600, res = 250)

pheatmap(
  top100,
  scale = "row",
  fontsize_row = 4,
  fontsize_col = 10,
  border_color = NA,
  annotation_col = annotation_col,
  annotation_colors = annotation_colors,
  main = "Top 100 most variable GO terms across mini-tank samples"
)

dev.off()
```

#KEGG temporal trends

```
library(ggplot2)


df <- data.frame(
  Timepoint = rep(c("T0", "T2", "T4", "T7"), times = 12),
  Tank = rep(c("MiniTank1", "MiniTank7"), each = 24),
  Pathway = rep(c(
    "map02024 Quorum sensing",
    "map02030 Bacterial chemotaxis",
    "map02020 Two-component system",
    "map00230 Purine metabolism",
    "map01230 Biosynthesis of amino acids",
    "map00500 Starch and sucrose metabolism"
  ), each = 4, times = 2),
  Value = c(
    # MiniTank1
    1.27423, 1.32477, 1.38600, 1.41830,   # map02024
    0.154758, 0.154859, 0.216973, 0.284140, # map02030
    1.06037, 1.05592, 1.18045, 1.36217,   # map02020
    2.58908, 2.52621, 2.43534, 2.34629,   # map00230
    3.19586, 3.17224, 3.18666, 3.11989,   # map01230
    1.06576, 0.96555, 0.915266, 0.821451, # map00500

    # MiniTank7
    1.23918, 1.31839, 1.34094, 1.38350,   # map02024
    0.227259, 0.170237, 0.210228, 0.259695, # map02030
    1.19606, 1.08633, 1.16949, 1.28902,   # map02020
    2.55203, 2.51273, 2.45111, 2.41257,   # map00230
    3.14300, 3.19100, 3.18409, 3.12453,   # map01230
    0.963602, 0.963488, 0.927986, 0.874766 # map00500
  )
)
```

```
df$Timepoint <- factor(df$Timepoint, levels = c("T0", "T2", "T4", "T7"))


p <- ggplot(df, aes(x = Timepoint, y = Value, group = Tank, colour = Tank)) +
  geom_line(linewidth = 0.9) +
  geom_point(size = 2) +
  facet_wrap(~ Pathway, scales = "free_y", ncol = 2) +
  labs(
    title = "Temporal trajectories of selected KEGG pathways",
    x = "Timepoint",
    y = "Relative abundance (%)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    strip.text = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    legend.position = "top"
  )

ggsave("KEGG_temporal_trends_selected.png", plot = p, width = 10, height = 12, dpi = 300)
```

#GO Temporal Trends

```
library(ggplot2)


# file paths
files <- c(
  "GO terms for all minitank1 time points.txt",
  "GO terms for all minitank7 timepoints.txt"
)


# selected GO terms
selected_terms <- c(
  "GO:0009058", # biosynthetic process
  "GO:0010467", # gene expression regulation
  "GO:0006412", # translation
  "GO:0016020", # membrane
  "GO:0050896", # response to stimulus
  "GO:0055114"  # oxidation-reduction process
)


# readable labels
term_labels <- c(
  "GO:0009058" = "GO:0009058 Biosynthetic process",
  "GO:0010467" = "GO:0010467 Gene expression regulation",
  "GO:0006412" = "GO:0006412 Translation",
  "GO:0016020" = "GO:0016020 Membrane",
  "GO:0050896" = "GO:0050896 Response to stimulus",
  "GO:0055114" = "GO:0055114 Oxidation-reduction process"
)
```

```
read_go_file <- function(file, tank_name) {
  lines <- readLines(file, warn = FALSE)

  current_time <- NA
  out <- data.frame()

  for (line in lines) {
    line <- trimws(line)

    if (line %in% c("T0:", "T2:", "T4:", "T7:")) {
      current_time <- sub(":", "", line)
      next
    }

    if (grepl("^[-0-9.eE]+\\s+GO:[0-9]+$", line) && !is.na(current_time)) {
      parts <- strsplit(line, "\\s+")[[1]]
      value <- as.numeric(parts[1])
      term <- parts[2]

      if (term %in% selected_terms) {
        out <- rbind(out, data.frame(
          Tank = tank_name,
          Timepoint = current_time,
          GO = term,
          Value = value
        ))
      }
    }
  }

  out
}
```

```
mt1 <- read_go_file(files[1], "MiniTank1")
mt7 <- read_go_file(files[2], "MiniTank7")

df <- rbind(mt1, mt7)
df$Timepoint <- factor(df$Timepoint, levels = c("T0", "T2", "T4", "T7"))
df$GO_label <- term_labels[df$GO]

p <- ggplot(df, aes(x = Timepoint, y = Value, group = Tank, colour = Tank)) +
  geom_line(linewidth = 0.9) +
  geom_point(size = 2) +
  facet_wrap(~ GO_label, scales = "free_y", ncol = 2) +
  labs(
    title = "Temporal trajectories of selected GO terms",
    x = "Timepoint",
    y = "Relative abundance (%)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    strip.text = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    legend.position = "top"
  )

ggsave("GO_temporal_trends_selected.png", plot = p, width = 10, height = 12, dpi = 300)
```

#KEGG Specific AMR Pathways

```
library(ggplot2)

df <- data.frame(
  Timepoint = rep(c("T0", "T2", "T4", "T7"), times = 6),
  Tank = rep(c("MiniTank1", "MiniTank7"), each = 12),
  Pathway = rep(c(
    "map01501 Beta-lactam resistance",
    "map01502 Vancomycin resistance",
    "map01503 CAMP resistance"
  ), each = 4, times = 2),
  Value = c(
    # MiniTank1
    0.475495, 0.426365, 0.429136, 0.422996,  # map01501
    0.262617, 0.253730, 0.246612, 0.235451,  # map01502
    0.267469, 0.254314, 0.260860, 0.290568,  # map01503

    # MiniTank7
    0.471442, 0.432841, 0.423380, 0.421235,  # map01501
    0.243552, 0.254079, 0.239178, 0.241041,  # map01502
    0.281686, 0.260926, 0.256503, 0.286153   # map01503
  )
)

df$Timepoint <- factor(df$Timepoint, levels = c("T0", "T2", "T4", "T7"))

p <- ggplot(df, aes(x = Timepoint, y = Value, group = Tank, colour = Tank)) +
  geom_line(linewidth = 0.9) +
  geom_point(size = 2) +
  facet_wrap(~ Pathway, scales = "free_y", ncol = 1) +
  labs(
    title = "Temporal trajectories of specific antibiotic resistance pathways",
    x = "Timepoint",
    y = "Relative abundance (%)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    strip.text = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    legend.position = "top"
  )

ggsave("KEGG_specific_AMR_pathways.png", plot = p, width = 8, height = 10, dpi = 300)
```

#GO Specific AMR Pathways

```
library(ggplot2)

files <- c(
  "GO terms for all minitank1 time points.txt",
  "GO terms for all minitank7 timepoints.txt"
)

selected_terms <- c(
  "GO:0046677", # response to antibiotic
  "GO:0006855", # drug transmembrane transport
  "GO:0017000"  # antibiotic binding
)

term_labels <- c(
  "GO:0046677" = "GO:0046677 Response to antibiotic",
  "GO:0006855" = "GO:0006855 Drug transmembrane transport",
  "GO:0017000" = "GO:0017000 Antibiotic binding"
)

read_go_file <- function(file, tank_name) {
  lines <- readLines(file, warn = FALSE)
  current_time <- NA
  out <- data.frame(
    Tank = character(),
    Timepoint = character(),
    GO = character(),
    Value = numeric(),
    stringsAsFactors = FALSE
  )
```

#GO Specific AMR Pathways

```
library(ggplot2)

files <- c(
  "GO terms for all minitank1 time points.txt",
  "GO terms for all minitank7 timepoints.txt"
)

selected_terms <- c(
  "GO:0046677", # response to antibiotic
  "GO:0006855", # drug transmembrane transport
  "GO:0017000"  # antibiotic binding
)

term_labels <- c(
  "GO:0046677" = "GO:0046677 Response to antibiotic",
  "GO:0006855" = "GO:0006855 Drug transmembrane transport",
  "GO:0017000" = "GO:0017000 Antibiotic binding"
)

read_go_file <- function(file, tank_name) {
  lines <- readLines(file, warn = FALSE)
  current_time <- NA
  out <- data.frame(
    Tank = character(),
    Timepoint = character(),
    GO = character(),
    Value = numeric(),
    stringsAsFactors = FALSE
  )
```

```
  for (line in lines) {
    line <- trimws(line)

    if (line %in% c("T0:", "T2:", "T4:", "T7:")) {
      current_time <- sub(":", "", line)
      next
    }

    if (grepl("^[-0-9.eE]+\\s+GO:[0-9]+$", line) && !is.na(current_time)) {
      parts <- strsplit(line, "\\s+")[[1]]
      value <- as.numeric(parts[1])
      term <- parts[2]

      if (term %in% selected_terms) {
        out <- rbind(out, data.frame(
          Tank = tank_name,
          Timepoint = current_time,
          GO = term,
          Value = value,
          stringsAsFactors = FALSE
        ))
      }
    }
  }

  out
}
```

```
mt1 <- read_go_file(files[1], "MiniTank1")
mt7 <- read_go_file(files[2], "MiniTank7")
df <- rbind(mt1, mt7)

# ensure all Tank x Timepoint x GO combinations exist
all_combos <- expand.grid(
  Tank = c("MiniTank1", "MiniTank7"),
  Timepoint = c("T0", "T2", "T4", "T7"),
  GO = selected_terms,
  stringsAsFactors = FALSE
)

df <- merge(all_combos, df, by = c("Tank", "Timepoint", "GO"), all.x = TRUE)

df$Timepoint <- factor(df$Timepoint, levels = c("T0", "T2", "T4", "T7"))
df$GO_label <- term_labels[df$GO]

p <- ggplot(df, aes(x = Timepoint, y = Value, group = Tank, colour = Tank)) +
  geom_line(linewidth = 1, na.rm = TRUE) +
  geom_point(size = 2.3, na.rm = TRUE) +
  facet_wrap(~ GO_label, scales = "free_y", ncol = 1) +
  labs(
    title = "Temporal trajectories of selected AMR-related GO terms",
    x = "Timepoint",
    y = "Relative abundance (%)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    strip.text = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    legend.position = "top"
  )

ggsave("GO_AMR_related_terms_final.png", plot = p, width = 8, height = 10, dpi = 300)
```